Jason Jaeseung Kim

Biomedical Engineering Undergraduate | Deep Learning for Medical Applications

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RESEARCH FOCUS

Through medical-imaging deep-learning research, I became interested not only in preprocessing and quantitative evaluation, but also in how model outputs can be interpreted and used with real clinical imaging data. Building on my experience with 3D MRI, I aim to extend this work to diverse clinical imaging problems.

EDUCATION

Yonsei University - B.S. Candidate in Biomedical Engineering *Mar 2021 - Expected Feb 2027*

Mirae Campus | Second Major in Electrical and Electronic Engineering, Sinchon Campus

GPA	4.50 Scale	Percentile	Department Rank
4.03 / 4.30	4.18 / 4.50	97.90 / 100	1 / 49

Selected coursework: Image Processing and Applications; Digital Signal Processing; Signals and Systems; Deep Learning Laboratory; Public Health and Medical Big Data Analysis.

RESEARCH EXPERIENCE

Former Undergraduate Researcher, Advanced Image Processing Techniques Lab, Yonsei University *Dec 2023 - Feb 2026*

Advisor: Prof. Joon Yul Choi | Mirae Campus

Deep Learning-Based Quantitative T1 Estimation | First-Author Manuscript in Preparation

- Designed comparative 3D MRI image-to-image regression experiments across input conditions, preprocessing and harmonization strategies, and model configurations; built automated pipelines for registration, masking, resampling, training, baseline reproduction, and tissue-wise evaluation.
- Contributed to drafting a multi-year government-funded R&D proposal and literature review; prepared the conference abstract and oral presentation; drafting the first-author manuscript, figures, and tables.

Quantitative MRI Study of Overwork | First-Author Manuscript in Preparation

- Designed and executed an SPM12 voxel-wise analysis, including cohort curation, covariate adjustment, explicit masking, FWE correction, atlas-based reporting, and automated QC and group summaries.
- Prepared the conference abstract, oral presentation, figures, and tables; drafting the first-author manuscript and coordinating revisions with the supervising professor.

Extending NAMER for Learning-Enhanced MRI Motion Estimation *Ongoing Graduation Research, 2026*

- NAMER estimates motion from a CNN-generated image prior before model-based reconstruction; current work evaluates multiple fastMRI subjects and revises CNN training so the prior better supports downstream motion estimation.

Undergraduate Researcher, Bio-Metamaterials Lab, Yonsei University *Jul 2022 - Aug 2022*

Advisor: Prof. Seung Ho Choi | Mirae Campus

- Participated in the initial installation and setup of an ellipsometer and analyzed the compositional and optical properties of biomaterials.
- Built a prototype tensile-strength measurement device and implemented a MATLAB-based user interface for measurement verification.

SELECTED RESEARCH OUTPUTS

- Oral Presentation and Outstanding Paper Award for the quantitative T1 estimation study, Korean Society of Medical and Biological Engineering Fall Conference, Nov 2025.
- Poster Presentation, NEURO2026, Jul 2026.

PROJECTS

Teacher-Difficulty-Calibrated Early-Exit Network *Undergraduate Course Project, 2026*

- Undergraduate Deep Learning Laboratory course project: developed an image-classification network using knowledge distillation, teacher-difficulty guidance, dynamic early exits, and branch-fused deployment for efficient inference.

Project Lead, CareProofRecord - Deep-Learning-Based Medical Explanation Video Platform *Jun 2025 - Apr 2026*

- Led the planning nce.ditirx rfchitecture of patient-facing medical explanation videos designed to improve understanding nce.reduce information-transfer errors.
- Designed a long-context generation workflow to mitigate hallucination risk, secured KRW 1,000,000 in university startup support, and explored clinical use with.regional medical institutions.

TECHNICAL SKILLS

Programming	Python; MATLAB
Medical Imaging Software	SynthStrip (skull stripping); FSL FAST (tissue segmentation); ANTs (registration nce.resampling); SPM12 (voxel-wise GLM and FWE correction); FreeSurfer (ROI extraction)

MERIT-BASED SCHOLARSHIPS

- National Science and Engineering Scholarship, Korea Student Aid Foundation - full tuition (Spring 2023, Fall 2023, Spring 2026).
- Hankyungbum Scholarship, Cheongpa-Hankyungbum Scholarship Foundation, Yonsei University - full tuition (Spring 2022, Fall 2022).
- Academic Excellence Scholarship, Yonsei University (Fall 2021).

HONORS AND AWARDS

- **Academic Excellence Award**, Yonsei University - Spring 2023
- **Academic Excellence Award**, Yonsei University - Fall 2022
- **Academic Honors Award**, Yonsei University - Fall 2021
- **Academic Excellence Award**, Yonsei University - Spring 2021
- **Grand Prize**, ASC Programming Contest - 2021
- **Excellence Award**, Wise Coding Life Algorithm Competition, Yonsei University - 2022
- **Grand Prize**, MEDICI+ Startup Competition, Yonsei University Wonju LINC+ Project Group - 2021
- **Honorable Mention**, WAVE-LAB Startup Competition - 2025
- **Encouragement Award**, Post Corona-19 Startup Challenge - 2022